최적제어 HW 1

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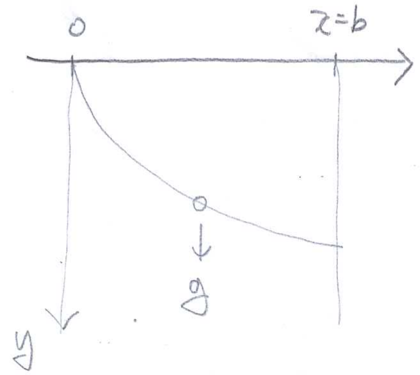

Point 0 : P_1 , Point b : P_2

total time : t_b , velocity : v , distance : s

$$t_b = \int_{P_1}^{P_2} \frac{ds}{v}$$

by Conservation of Energy

$$\frac{1}{2} m v^2 = mgy \rightarrow v = \sqrt{2gy}$$

$$ds = \sqrt{dx^2 + dy^2} = \sqrt{1+y'^2}\, dx$$

$$t_b = \int_{P_1}^{P_2} \frac{\sqrt{1+y'^2}}{\sqrt{2gy}}\, dx = \int_{P_1}^{P_2} \sqrt{\frac{1+y'^2}{2gy}}\, dx$$

Let $f = (1+y'^2)^{1/2} (2gy)^{-1/2}$

Euler-Lagrange differential equation

$$\frac{\partial f}{\partial y} - \frac{d}{dx}\left(\frac{\partial f}{\partial y'}\right) = 0 \quad \text{↳ Beltrami Identity}$$

$$f - y' \frac{\partial f}{\partial y'} = C \quad -①$$

$$\frac{\partial f}{\partial y'} = \frac{y'}{\sqrt{1+y'^2}\sqrt{2gy}}$$

①에 대입 → $\dfrac{\sqrt{1+y'^2}}{\sqrt{2gy}} - y' \dfrac{y'}{\sqrt{2gy}\sqrt{1+y'^2}} = \dfrac{1+y'^2-y'^2}{\sqrt{2gy}\sqrt{1+y'^2}}$

$$= \frac{1}{\sqrt{2gy}\sqrt{1+y'^2}}$$

$$\frac{1}{\sqrt{2gy}\sqrt{1+y'^2}} = \text{constant}$$

$\Rightarrow \left[1 + \left(\frac{dy}{dx}\right)^2\right] y = C$ ↳ 뒷면에서

$\Rightarrow$ sol : $x = \frac{C}{2}(\theta - \sin\theta)$

$y = \frac{C}{2}(1-\cos\theta)$

BC $\theta = \frac{\pi}{2}$ $\left[\begin{array}{l} x = b \rightarrow \frac{C}{2}(\frac{\pi}{2} - 1) = b \quad \therefore C = \dfrac{2b}{\frac{\pi}{2}-1} \\ y = \frac{C}{2} = \dfrac{b}{\frac{\pi}{2}-1} \end{array}\right.$

$\therefore \left[\begin{array}{l} x = \dfrac{b}{\frac{\pi}{2}-1}(\theta - \sin\theta) \\ y = \dfrac{b}{\frac{\pi}{2}-1}(1-\cos\theta) \end{array}\right.$ $\left(0 \le \theta \le \frac{\pi}{2}\right)$

$$\left[1+\left(\frac{dy}{dx}\right)^2\right] y = C$$

let $\frac{dy}{dx} = \tan\theta$

$1+\tan^2\theta = \sec^2\theta$

$\sec^2\theta \, y = C \rightarrow y = c\cos^2\theta$

$dy = 2C \cdot (\cos\theta)(-\sin\theta)\, d\theta$

$\frac{dy}{d\theta} = -2c\cos\theta\sin\theta$ — ①

$$\frac{dx}{d\theta} = \frac{dx}{dy}\frac{dy}{d\theta} = \frac{\cos\theta}{\cancel{\sin\theta}}(-2c\cos\theta\cancel{\sin\theta})$$

$$= -2c\cos^2\theta \quad \text{—②}$$

② 적분을 위해 Let $\varphi = 2\theta$

$$\frac{dx}{d\theta} = -2c \cos^2\frac{\varphi}{2} = (-2c)\frac{1+\cos\varphi}{2}$$

$$\int dx = -c(1+\cos\varphi)\, d\theta \qquad (d\theta = \tfrac{1}{2} d\varphi)$$

$$= \int -\frac{c}{2}(1+\cos\varphi)\, d\varphi$$

$$x = -\frac{c}{2}(\varphi + \sin\varphi) \quad \text{—③}$$

① 에서 $\int dy = \int -2c\cos\theta\sin\theta \, d\theta$

$$y = -C\sin^2\theta = \frac{-c}{2}(1-\cos\varphi). \quad \text{—④}$$

③, ④ 에서 $c' = -c$ 라고 하면 "sol"이 된다.